

Class Factorized Variational Auto-encoder for Radar HRRP Target Recognition

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Abstract—In this paper, a class factorized variational auto-encoder (CFVAE) is developed for radar high-resolution range profile (HRRP) target recognition. The proposed model integrates the discriminative information into the deep statistical modeling of HRRP. In detail, the CFVAE utilizes an encoder to project all observations to the deep latent space and then separately defines the generative process from the latent features to observed data in each class, via the collection of specific class-decoders. Due to the stronger ability of each class-decoder to the description of observations in the corresponding class, the CFVAE model can directly assign a test sample into the class that corresponds to the decoder with the minimum reconstruction error, which avoids the mismatch of the extracted features and the back-end classifier. Meanwhile, compared to the traditional variational auto-encoder (VAE) describing the whole dataset with a single decoder, the proposed method has the capacity to give a more accurate description for all observations, thus beneficial to the improvement of features in characterization ability. Moreover, instead of imposing a fixed prior on the latent representations, our CFVAE model learns the conditional prior distribution based on the samples' labels, which further enhances the discrimination of the latent space. Finally, experiments on the measured HRRP dataset demonstrate the promising recognition performance of the proposed method.

Keywords—High-resolution range profile (HRRP), radar automatic target recognition (RATR), variational auto-encoder (VAE), class factorizing, minimum reconstruction error

I. INTRODUCTION

Radar high-resolution range profile (HRRP) is the amplitude of the coherent summations of complex returns from target scatterers in each range cell. Since it contains abundant target information and has the properties of easy acquisition and processing, the radar automatic target recognition (RATR) based on HRRP has received extensive attention [1]–[6], [12]–[15].

In HRRP-based RATR, the quality of the data feature representation has significant effect on the performance of the recognition system. Feng et al. [1] utilize the K-step singular value decomposition (K-SVD) algorithm to learn the sparse overcomplete representation of the HRRP. In [2], the principal component analysis (PCA) is adopted to represent the HRRP with some linear uncorrelated principal components. Moreover, Du et al. [3] apply the factor analysis (FA) model to the learning of low-dimensional latent factors hidden in HRRP data.

Although above methods have been proved to be effective, they are shallow models and fail to learn the deep representations of the observed data, which is more

advantageous in various real-world tasks [4]–[6], [16]–[18]. In combination of the merits of the deep network and probabilistic model, a variational auto-encoder (VAE) is proposed in [7] and realizes the in-depth feature extraction. The VAE models observations via a multi-layered decoder and meanwhile approximates the posterior distribution of the latent feature by using an encoder. Due to the good expressive power and flexible scalability, the VAE is used in many applications [8]–[11].

However, the VAE is an unsupervised model which cannot be directly used for radar HRRP target recognition. To tackle this problem, we introduce the label information of the observed data to the traditional VAE and propose a discriminative deep network, i.e., class factorized VAE (CFVAE). In the CFVAE, the decoding module partitions the whole dataset into several parts according to observations' class labels, and the observed data in each part is depicted with a specific class-decoder. Since the observed samples from different classes are exploited to train the different decoders, each class-decoder can only accurately describe the observations from the corresponding class. Based on this property, the CFVAE can be directly used for the target recognition task with the criteria of minimum reconstruction error, thus realizing the joint learning of the latent feature and classifier. Moreover, compared to the traditional VAE which only contains a single decoder, the decoding manner according to data category in the CFVAE allows each class-decoder to specialize in a smaller part of a larger data description problem. Due to the fact that a single decoder can better fit the observations with small size and simple probability distribution, the collection of decoders offers the potential to make a more precise estimation for the large-scale dataset. Therefore, the CFVAE model can more accurately depict the whole observed dataset than the traditional VAE, which avoids the weak representation ability of deep features caused by the inaccurate depiction of observations.

Note that the feature prior in the VAE-based models carries our understanding of the underlying feature distribution and determines how the learned latent space is structured, which is crucial to the performance of the model. Rather than a fixed standard Gaussian prior for all observed data in the traditional VAE, another contribution of the proposed CFVAE is that the prior of the latent feature is learned via a network conditioned on the label information. The differences among class labels can enhance the separability of the feature prior from different classes and facilitate the learning of a latent space with stronger discrimination, thus leading to the further improvement of the proposed model in recognition performance.

The CFVAE model can ultimately be implemented by maximizing the lower bound of the log marginal likelihood

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TABLE I. LIST OF NOTATIONS IN THE CFVAE MODEL

Notation	Meaning	Notation	Meaning
N	The number of observed samples	$\boldsymbol{\mu}_\varphi(\mathbf{x}_n)$	Mean vector of $\mathbf{z}_n$
C	The number of classes	$\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)$	Standard deviation vector of $\mathbf{z}_n$
L	The sampling number of the latent feature	$\boldsymbol{\mu}_{\theta_{c_n}}(\mathbf{z}_n)$	Mean vector of $\mathbf{x}_n$
$\mathbf{x}_n$	The n th observed sample with $\mathbf{X} = \{\mathbf{x}_n\}_{n=1}^N$	$\boldsymbol{\sigma}_{\theta_{c_n}}(\mathbf{z}_n)$	Standard deviation vector of $\mathbf{x}_n$
$\mathbf{z}_n$	Latent feature	$\boldsymbol{\mu}_\phi(\mathbf{y}_n)$	Prior mean vector of $\mathbf{z}_n$
$\mathbf{y}_n$	The class label of $\mathbf{x}_n$	$\boldsymbol{\sigma}_\phi(\mathbf{y}_n)$	Prior standard deviation vector of $\mathbf{z}_n$
$\boldsymbol{\varepsilon}_n$	Auxiliary variable with $\boldsymbol{\varepsilon}_n \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$	$\mathbf{W}_1, \dots, \mathbf{W}_6$	Weight matrices of the network
c_n	Class indicator of $\mathbf{x}_n$ with $c_n \in \{1, \dots, C\}$ and $\mathbf{c} = \{c_n\}_{n=1}^N$	$\mathbf{b}_1, \dots, \mathbf{b}_6$	Biases of the network
θ_c	Parameter of the decoder c with $\boldsymbol{\theta} = \{\theta_c\}_{c=1}^C$	$\mathcal{N}(\cdot)$	Gaussian distribution
φ	Parameter of the encoder	$\text{sigm}(\cdot)$	Sigmoid function
ϕ	Parameter of the prior network	$\text{KL}(\cdot)$	Kullback-Leibler divergence

of the observations. In the test phase, the generated samples of a test sample under each class-decoder can be obtained via the feed-forward propagation of the trained CFVAE, and then according to the minimum reconstruction error between the original and generated samples, we can eventually determine the class membership of the test HRRP sample.

In Table I, we list all notations in our CFVAE model to help readers to understand the dimensions of matrices and definitions of notations clearly.

In the remaining paper, Section 2 gives a brief review of the traditional VAE. A detailed introduction to the CFVAE is provided in Section 3. Section 4 conducts the experiments on the measured HRRP dataset to evaluate the performance of the proposed method, and Section 5 summarizes the discussion.

II. VARIATIONAL AUTO-ENCODER

The VAE is known as a probabilistic generative model [7], [11], which has the capacity to reveal the deep latent representations of data. Given the observed dataset $\mathbf{X} = \{\mathbf{x}_n\}_{n=1}^N$ with N denoting the number of observed samples, the VAE depicts the generative process from latent feature $\mathbf{z}_n$ to $\mathbf{x}_n$ via a decoder with parameter θ , i.e., $p_\theta(\mathbf{x}_n|\mathbf{z}_n)$. For the strong expression ability, the decoder usually adopts a deep neural network, such as the multilayer perception (MLP), and outputs the distribution parameter of $p_\theta(\mathbf{x}_n|\mathbf{z}_n)$, such as the mean $\boldsymbol{\mu}_\theta(\mathbf{z}_n)$ and standard

deviation $\boldsymbol{\sigma}_\theta(\mathbf{z}_n)$ of Gaussian distribution. The goal of the VAE is to maximize the lower bound $\sum_{n=1}^N \text{LB}(\theta, \varphi; \mathbf{x}_n)$ of the log marginal likelihood $\ln p_\theta(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$ via the stochastic gradient ascent (SGA) algorithm [19], with

$$\text{LB}(\theta, \varphi; \mathbf{x}_n) = \mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_\theta(\mathbf{x}_n|\mathbf{z}_n)] - \text{KL}(q_\varphi(\mathbf{z}_n|\mathbf{x}_n) \| p_\theta(\mathbf{z}_n)) \quad (1)$$

The first term of right-hand side in (1) is the expectation of log likelihood $\ln p_\theta(\mathbf{x}_n|\mathbf{z}_n)$ with respect to the variational distribution $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$. $p_\theta(\mathbf{z}_n)$ is the prior distribution of the latent feature $\mathbf{z}_n$. Due to the complex form of $\mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_\theta(\mathbf{x}_n|\mathbf{z}_n)]$, the solution of the lower bound is often non-analytical. Alternatively, the VAE represents $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$ by utilizing an encoder parameterized with φ , which takes $\mathbf{x}_n$ as the input and outputs the distribution parameters of $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$. And then, with the Monte Carlo sampling, $\mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_\theta(\mathbf{x}_n|\mathbf{z}_n)]$ has the explicit expression of $L^{-1} \sum_{l=1}^L \ln p_\theta(\mathbf{x}_n|\mathbf{z}_n^l)$ with $\mathbf{z}_n^l$ being the l th sampling value of $\mathbf{z}_n$ sampled from $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$. However, the sampling of $\mathbf{z}_n$ hinders the back-propagation of error. To solve this problem, the VAE introduces an auxiliary variable $\boldsymbol{\varepsilon}_n$ and reparameterizes $\mathbf{z}_n$ as $\mathbf{z}_n = \mathbf{g}_\varphi(\mathbf{x}_n, \boldsymbol{\varepsilon}_n)$ with $\boldsymbol{\varepsilon}_n \sim p(\boldsymbol{\varepsilon}_n)$. In this paper, the Gaussian distribution is adopted for $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$ with its mean $\boldsymbol{\mu}_\varphi(\mathbf{x}_n)$ and standard deviation $\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)$ being encoded by an encoder. Based on the reparameterization trick, we can represent $\mathbf{z}_n$ as

$$\mathbf{z}_n = \mathbf{g}_\varphi(\mathbf{x}_n, \boldsymbol{\varepsilon}_n) = \boldsymbol{\mu}_\varphi(\mathbf{x}_n) + \boldsymbol{\sigma}_\varphi(\mathbf{x}_n) \odot \boldsymbol{\varepsilon}_n, \quad \boldsymbol{\varepsilon}_n \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (2)$$

where $\mathcal{N}(\cdot)$ denotes the Gaussian distribution, and $\odot$ denotes the element-wise product operator. Finally, we can rewrite the lower bound $\text{LB}(\theta, \varphi; \mathbf{x}_n)$ as

$$\text{LB}(\theta, \varphi; \mathbf{x}_n) \simeq \frac{1}{L} \sum_{l=1}^L \ln p_\theta(\mathbf{x}_n|\mathbf{z}_n^l) - \text{KL}(q_\varphi(\mathbf{z}_n|\mathbf{x}_n) \| p_\theta(\mathbf{z}_n)) \quad (3)$$

with $\mathbf{z}_n^l = \boldsymbol{\mu}_\varphi(\mathbf{x}_n) + \boldsymbol{\sigma}_\varphi(\mathbf{x}_n) \odot \boldsymbol{\varepsilon}_n^l$, and $\boldsymbol{\varepsilon}_n^l$ is the l th sampling value of $\boldsymbol{\varepsilon}_n$.

Usually, the VAE imposes a standard Gaussian prior on $\mathbf{z}_n$, i.e., $p_\theta(\mathbf{z}_n) = \mathcal{N}(\mathbf{z}_n|\mathbf{0}, \mathbf{I})$. Then the Kullback-Leibler (KL) divergence between $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$ and $p_\theta(\mathbf{z}_n)$ has the explicit expression of

$$\text{KL}(q_\varphi(\mathbf{z}_n|\mathbf{x}_n) \| p_\theta(\mathbf{z}_n)) = -\frac{1}{2} \sum_{j=1}^J \left[1 + \ln(\sigma_\varphi^{(n,j)})^2 - (\mu_\varphi^{(n,j)})^2 - (\sigma_\varphi^{(n,j)})^2 \right] \quad (4)$$

where $\mu_\varphi^{(n,j)}$ and $\sigma_\varphi^{(n,j)}$ are the j th elements of $\boldsymbol{\mu}_\varphi(\mathbf{x}_n)$ and $\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)$, respectively. By combining (2) and (4), the lower bound in (3) can be optimized via the SGA algorithm. For

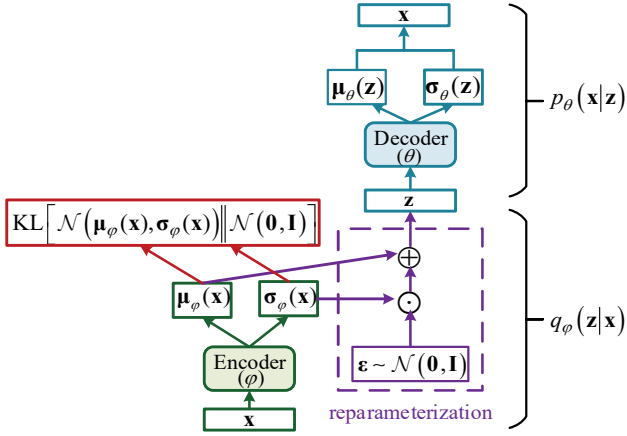


Fig. 1. Architecture of the VAE with Gaussian distribution assumption.

more intuitive explanation, we take the Gaussian distribution as an example and present the structure of VAE in Fig. 1.

III. THE PROPOSED METHOD

A. Class Factorized Variational Auto-encoder

Due to the unsupervised learning of the VAE, it can not directly used for the HRRP-based target recognition. In this paper, we introduce the class labels of HRRP samples to the VAE and develop a discriminative deep model, i.e., CFVAE. In detail, the decoding network of the CFVAE is factorized into several sub-networks according to the class information of observations. Each sub-network individually describes the generative process of the data from the corresponding class via a specific decoder.

Suppose that the whole observed HRRP dataset $\mathbf{X} = \{\mathbf{x}_n, n=1, \dots, N\}$ covers C classes and each sample $\mathbf{x}_n$ occupies a class indicator $c_n \in \{1, \dots, C\}$. The goal of CFVAE is to maximize the log likelihood of observations given the class information, i.e.,

$$\begin{aligned} \ln p_{\theta}(\mathbf{X}|\mathbf{c}) &\geq \text{LB}(\theta, \varphi; \mathbf{X}, \mathbf{c}) \\ &= \sum_{n=1}^N \mathbb{E}_{q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_{\theta}(\mathbf{x}_n|\mathbf{z}_n, c_n)] - \text{KL}(q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n) \| p_{\theta}(\mathbf{z}_n|c_n)) \quad (5) \\ &= \sum_{n=1}^N \mathbb{E}_{q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)] - \text{KL}(q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n) \| p_{\theta}(\mathbf{z}_n|c_n)) \end{aligned}$$

where $\mathbf{c} = \{c_n\}_{n=1}^N$, $\theta = \{\theta_c\}_{c=1}^C$ with θ_c being the parameter of the decoder corresponding to the class c . The expression $\mathbb{E}_{q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)]$ in (5) indicates that each observed sample $\mathbf{x}_n$ ($n=1, \dots, N$) is generated from the mapping of the latent feature $\mathbf{z}_n$ via a specified class-decoder with parameter θ_{c_n} , and the posteriors of all latent features $\{\mathbf{z}_n\}_{n=1}^N$ are approximated via an encoder. As a result, the optimization of the decoding parameter θ_c just depends on those samples belonging to the c th class, which makes each decoder performs well only for the observations from the corresponding class.

In addition, different from the traditional VAE, the CFVAE encourages the posterior of the latent feature close to the prior distribution conditioned on the class information,

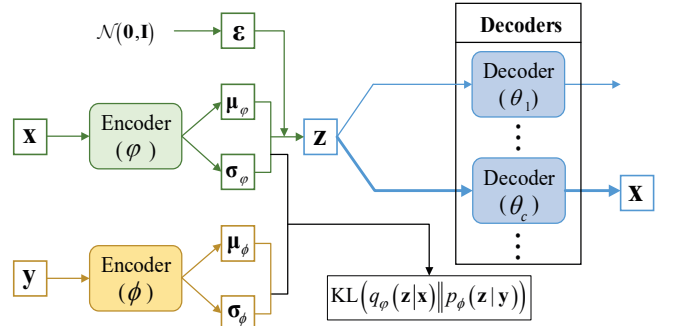


Fig. 2. Architecture of the CFVAE model. The sample $\mathbf{x}$ is projected into the latent space via an encoder with parameter φ , and then the latent variable $\mathbf{z}$ is fed to the decoder with parameter θ_c according to its class indicator. The prior of the latent variable $\mathbf{z}$ is learned via a network with parameter ϕ conditioned on its label $\mathbf{y}$, and the KL term pushes the posterior $q_{\varphi}(\mathbf{z}|\mathbf{x})$ close to the prior $p_{\phi}(\mathbf{z}|\mathbf{y})$.

i.e., $\text{KL}(q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n) \| p_{\phi}(\mathbf{z}_n|c_n))$ in (5). Therefore, instead of giving a fixed standard Gaussian distribution, we learn the conditional prior distribution for $\mathbf{z}_n$ conditioned on its class label via an extra network, which can be represented as

$$p_{\theta}(\mathbf{z}_n|c_n) \triangleq p_{\phi}(\mathbf{z}_n|\mathbf{y}_n) \quad (6)$$

with $\mathbf{y}_n \in \mathbb{R}^{C \times 1}$ denoting the class label of $\mathbf{x}_n$ and usually being set to a one-hot vector, ϕ being the parameter of the extra prior network. Furthermore, the lower bound $\text{LB}(\theta, \varphi; \mathbf{X}, \mathbf{c})$ can be expanded as

$$\begin{aligned} \text{LB}(\theta, \varphi; \mathbf{X}, \mathbf{c}) &= \sum_{n=1}^N \mathbb{E}_{q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)] \\ &\quad - \text{KL}(q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n) \| p_{\phi}(\mathbf{z}_n|\mathbf{y}_n)) \quad (7) \end{aligned}$$

Due to the differences among class labels, the learned conditional prior distribution from different classes is more distinguishable. With the KL constraint, the learned latent features can thus have the stronger inter-class separability, which is propitious to the further improvement in model recognition performance.

In radar HRRP target recognition, as a compromise between accuracy and computation resource, the Gaussian distribution is widely used for the description of HRRP [3]. Thus, we set $p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)$ to

$$p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n) = \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_{\theta_{c_n}}(\mathbf{z}_n), \text{diag}(\boldsymbol{\sigma}_{\theta_{c_n}}^2(\mathbf{z}_n))), \quad (8)$$

where $\boldsymbol{\mu}_{\theta_{c_n}}(\mathbf{z}_n)$ and $\boldsymbol{\sigma}_{\theta_{c_n}}(\mathbf{z}_n)$ represent the mean and standard deviation vectors outputted from the decoder with parameter θ_{c_n} , $\text{diag}(\cdot)$ denotes the diagonalization operation.

Analogously, the Gaussian posterior $q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n)$ can be expressed as

$$q_{\varphi}(\mathbf{z}_n|\mathbf{x}_n) = \mathcal{N}(\mathbf{z}_n | \boldsymbol{\mu}_{\varphi}(\mathbf{x}_n), \text{diag}(\boldsymbol{\sigma}_{\varphi}^2(\mathbf{x}_n))), \quad (9)$$

with the mean $\boldsymbol{\mu}_{\varphi}(\mathbf{x}_n)$ and standard deviation $\boldsymbol{\sigma}_{\varphi}(\mathbf{x}_n)$ being outputted from the encoder. Moreover, the conditional prior over the latent features $p_{\phi}(\mathbf{z}_n|\mathbf{y}_n)$ can be expressed as

$$p_{\phi}(\mathbf{z}_n|\mathbf{y}_n) = \mathcal{N}(\mathbf{z}_n | \boldsymbol{\mu}_{\phi}(\mathbf{y}_n), \text{diag}(\boldsymbol{\sigma}_{\phi}^2(\mathbf{y}_n))), \quad (10)$$

and then, the KL divergence in (7) can be analytically

represented as

$$\begin{aligned} \text{KL}(q_\phi(\mathbf{z}_n | \mathbf{x}_n) \| p_\phi(\mathbf{z}_n | \mathbf{y}_n)) = \\ \frac{1}{2} \sum_{n=1}^N \sum_{j=1}^J \left[1 + \ln(\sigma_\phi^{(n,j)})^2 - \ln(\sigma_\phi^{(n,j)})^2 \right] - \frac{1}{2} \sum_{n=1}^N \sum_{j=1}^J \left[(\sigma_\phi^{(n,j)})^{-2} \times \right. \\ \left. \left((\mu_\phi^{(n,j)})^2 + (\sigma_\phi^{(n,j)})^2 - 2\mu_\phi^{(n,j)}\mu_\phi^{(n,j)} + (\mu_\phi^{(n,j)})^2 \right) \right] \end{aligned} \quad (11)$$

Combining with (8), (11), and the reparameterization, the lower bound $\text{LB}(\theta, \phi; \mathbf{X}, \mathbf{c})$ in the CFVAE has the explicit expression and can be optimized via the SGA algorithm. In Fig. 2, we give the network structure of the CFVAE.

B. Network Implementation

In the CFVAE, we adopt the MLP architecture to construct the encoder, decoders, and prior network. The layer number of the MLP should be adjusted according to the data size. Here, we take an example of MLP with a single hidden layer to demonstrate the derivations of the latent feature $q_\phi(\mathbf{z}|\mathbf{x})$, observed data $p_{\theta_c}(\mathbf{x}|\mathbf{z})$, and the conditional prior $p_\phi(\mathbf{z}|\mathbf{y})$.

In detail, the posterior distribution of the latent feature $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}_\phi(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x})))$ can be derived via the following way:

$$\mathbf{h}_1 = \text{sigm}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1), \quad (12)$$

$$\boldsymbol{\mu}_\phi(\mathbf{x}) = \mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2, \quad \boldsymbol{\sigma}_\phi(\mathbf{x}) = \text{sigm}(\mathbf{W}_3 \mathbf{h}_1 + \mathbf{b}_3) \quad (13)$$

where $\phi = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \mathbf{W}_3, \mathbf{b}_3\}$ with $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{W}_3$ denoting the weight matrices and $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$ the biases, respectively; $\text{sigm}(\cdot)$ denotes the sigmoid function. Similarly to the posterior distribution of $\mathbf{z}$, the conditional distribution of observed sample $\mathbf{x}$ given its class label c , i.e., $p_{\theta_c}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_{\theta_c}(\mathbf{z}), \text{diag}(\boldsymbol{\sigma}_{\theta_c}^2(\mathbf{z})))$, can be expressed as

$$\mathbf{h}_2 = \text{sigm}(\mathbf{W}_4 \mathbf{z} + \mathbf{b}_4), \quad (14)$$

$$\boldsymbol{\mu}_{\theta_c}(\mathbf{z}) = \mathbf{W}_5 \mathbf{h}_2 + \mathbf{b}_5, \quad \boldsymbol{\sigma}_{\theta_c}(\mathbf{z}) = \text{sigm}(\mathbf{W}_6 \mathbf{h}_2 + \mathbf{b}_6), \quad (15)$$

where θ_c denotes the parameter of the decoder selected by $\mathbf{x}$ according to its class information and $\theta_c = \{\mathbf{W}_4, \mathbf{b}_4, \mathbf{W}_5, \mathbf{b}_5, \mathbf{W}_6, \mathbf{b}_6\}$. Moreover, the conditional prior $p_\phi(\mathbf{z}|\mathbf{y}) = \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}_\phi(\mathbf{y}), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{y})))$ has the form of

$$\mathbf{h}_3 = \text{sigm}(\mathbf{W}_7 \mathbf{y} + \mathbf{b}_7), \quad (16)$$

$$\boldsymbol{\mu}_\phi(\mathbf{y}) = \mathbf{W}_8 \mathbf{h}_3 + \mathbf{b}_8, \quad \boldsymbol{\sigma}_\phi(\mathbf{y}) = \text{sigm}(\mathbf{W}_9 \mathbf{h}_3 + \mathbf{b}_9), \quad (17)$$

with $\phi = \{\mathbf{W}_7, \mathbf{b}_7, \mathbf{W}_8, \mathbf{b}_8, \mathbf{W}_9, \mathbf{b}_9\}$. Based on SGA, the solutions of ϕ , θ and θ can be obtained. After the model learning, the reconstruction errors of a test sample $\mathbf{x}^*$ given all class-decoders can be calculated by going through the feed-forward propagation of the learned CFVAE, and then $\mathbf{x}^*$ can be classified to the class corresponding to the decoder with the minimum reconstruction error.

IV. EXPERIMENTS

A. Data Description and Experimental Setup

We examine the performance of our CFVAE model on

the measured HRRP dataset from three real airplanes, including Yark-42, Cessna Citation S/II, and An-26 [1]-[6], [14], [15], with the radar center frequency of 5520MHz and the bandwidth of 400MHz. We select 7000 HRRPs from the Yark-42, 10000 HRRPs from the Cessna Citation S/II, 10000 HRRPs from the An-26 as the training samples, and other 5200 HRRPs from three targets as the test samples.

The encoding MLP architecture in the CFVAE is set to 256-1500-500-50 with “256” denoting the dimension of HRRP sample, and the symmetrical structures are adopted for all three decoders, i.e., $C=3$. The prior network architecture is set to 3-10-30-50. The network weights are initialized by randomly sampling from $\mathcal{N}(0,1)$, and the bias are initialized as 0. The learning rate of the SGA algorithm is set to 0.001.

B. Performance Analysis

To demonstrate the descriptive ability of the class-decoders in the CFVAE, we train the proposed model with 27000 HRRPs and then feed the test HRRP data into the trained CFVAE to generate their reconstructed samples. The reconstructed HRRPs via three different class-decoders for a test sample randomly selected from each target are shown in Fig. 3, in which different rows from top to bottom represent the reconstructions of the test HRRPs from An-26, Cessna Citation S/II and Yark-42, respectively, and the columns from left to right represent the reconstructions via the different class-decoders respectively corresponding to An-26, Cessna Citation S/II and Yark-42 (referred to as the decoder 1, decoder 2, and decoder 3 for convenience).

From Fig. 3 we note that each class-decoder can better reconstructs those HRRPs from the corresponding class, as presented in Figs. 3(a), (e) and (i); on the contrary, the class-decoder has the poor reconstruction capability to data from other classes. Taking the test HRRP from Yark-42 as an example, the reconstructions of signals from the 100th to 150th range cells via the decoder 3 are obviously better than those via the decoder 1 and decoder 2, as shown in Figs. 3(g), (h) and (i). Therefore, it is reasonable to determine the class membership of a test HRRP sample with the minimum reconstruction error. For a clearer illustration, Table II gives a comparison to the mean square errors (MSEs) of all test HRRPs from each class under different class-decoders, from which the same conclusions as the Fig. 3 can be obtained.

In the following, we validate the recognition performance of the proposed method by comparing with some shallow models including K-SVD [1], PCA [2], FA [3], and several deep networks including traditional VAE, stacked corrective auto-encoder (SCAE) [4], target-aware recurrent attentional network (TARAN) [5], factorized discriminative conditional variational auto-encoder (FDCVAE) [6]. The traditional

TABLE II. MSEs OF ALL TEST HRRP SAMPLES FROM EACH CLASS UNDER DIFFERENT CLASS-DECODERS OF THE CFVAE MODEL

Decoders Class of the test HRRPs	Decoder 1	Decoder 2	Decoder 3
An-26	0.3123	0.5307	0.4614
Cessna Citation S/II	0.3958	0.3076	0.4523
Yark-42	0.4233	0.6641	0.3321

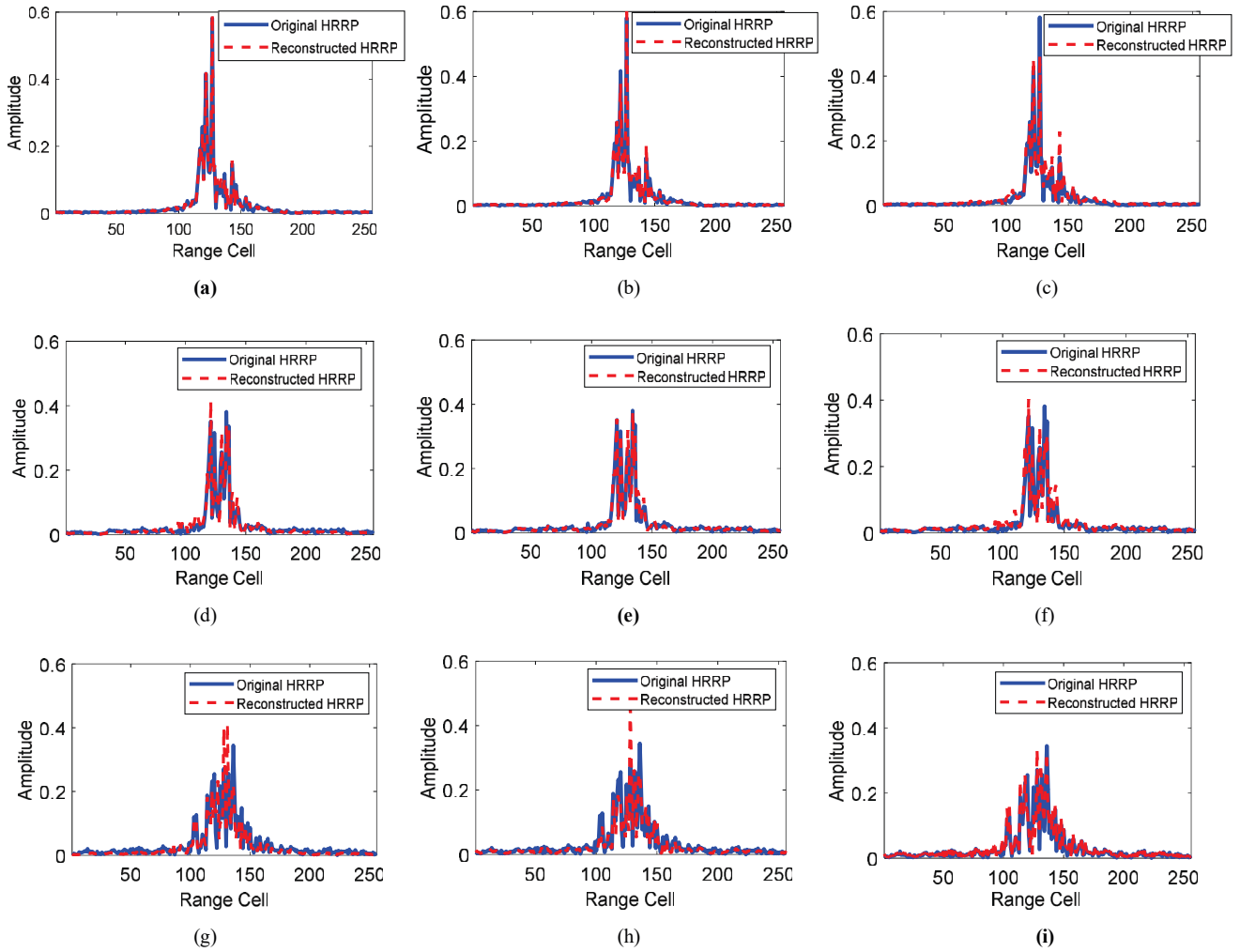


Fig. 3 Reconstructions of the test HRRP samples via three different class-decoders of the CFVAE. Different rows from top to bottom represent the reconstructions of different test HRRP samples respectively from An-26, Cessna Citation S/II and Yark-42; the different columns from left to right represent the reconstructions with the different class-decoders that correspond to An-26, Cessna Citation S/II and Yark-42, respectively.

TABLE III. RECOGNITION PERFORMANCE OF THE PROPOSED MODELS AND RELATED METHODS.

Methods	Accuracy (%)
PCA	81.44
K-SVD	85.32
FA	85.51
VAE	86.08
SCAE	88.04
TARAN	88.25
FDCVAE	88.43
CFVAE	90.01

VAE means that the latent feature is firstly captured via the VAE model, and then the extracted features are used to learn the linear support machine (LSVM). The SCAE, TARAN, and FDCVAE models are specifically built for the modeling of HRRP with FDCVAE also being the extension of the VAE.

Table III presents the recognition accuracies of our CFVAE and other related models. As shown in Table III, the deep networks exhibit stronger recognition ability than the shallow models, due to their characteristic of exploring

more effective features. Among deep models, VAE is unsupervised and there may be a mismatch of the extracted feature and classifier, thus having the lower recognition rates than the other deep models. Moreover, since the probabilistic models can learn the feature with the better representation capability, the recognition performance of the CFVAE and FDCVAE is superior to that of the SCAE and TARAN, which are constructed under the algebraic frame. Finally, rather than building an extra classification

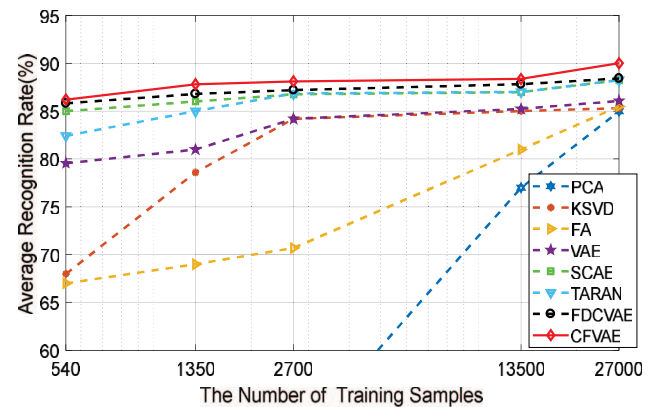


Fig. 4. Recognition performance of the CFVAE and other related models with different training data sizes.

network in the FDCVAE, the recognition scheme based on the minimum reconstruction error avoids the “incompatibility” between the recognition and reconstruction networks. Therefore, our CFVAE model achieves the best recognition performance.

Due to the limitation of the radar sampling rate, it usually is hard to obtain enough training HRRP samples in the real radar systems. To investigate the generalization ability of the CFVAE under the limited training data condition, we show the recognition performance of above models with different training data sizes in Fig. 4, from which we see that the recognition performance of shallow models dramatically degrades when the number of training samples decreases. This is because a shallow model is constructed for each aspect-frame of each target, which requires abundant training data in each frame. However, the deep methods share a model among all training HRRPs from all targets. The parameter sharing greatly reduces the model complexity, which leads to the more robust recognition performance with small training data size. Finally, the probabilistic deep methods (the CFVAE and FDCVAE) can describe the uncertainty of data, thus having the stronger generalization ability.

V. CONCLUSION

This paper presents a novel probabilistic framework named CFVAE for radar HRRP target recognition. To explore the discriminative deep features of the HRRPs, the CFVAE introduces the class information into the VAE by using the multiple class-decoders to separately learn the generative process of the observed data from each class. Meanwhile, due to the stronger description capability of the single decoder to a small patch of the observed data, the CFVAE is capable of giving a more precise depiction to all observations via the collection of class-decoders. In addition, the conditional prior given class labels for the latent features further promotes the learning of the discriminative latent space. Experiments on the measured HRRP data demonstrate the excellent recognition performance of the proposed CFVAE.

REFERENCES

- [1] B. Feng, L. Du, H.W. Liu, and F. Li, “Radar HRRP target recognition based on K-SVD algorithm,” *Proceedings of the IEEE CIE International Conference on Radar*, pp. 642–645, October 2011.
- [2] L. Du, H.W. Liu, Z. Bao, and J. Zhang, “Radar automatic target recognition using complex high-resolution range profiles,” *IET Radar Sonar, and Navigation*, vol. 1, no. 1, pp. 18–26, March 2007.
- [3] L. Du, H.W. Liu, and Z. Bao, “Radar HRRP statistical recognition: Parametric model and model selection,” *IEEE Transactions on Signal Processing*, vol. 56, no. 5, pp. 1931–1944, April 2008.
- [4] B. Feng, B. Chen, and H.W. Liu, “Radar HRRP target recognition with deep networks,” *Pattern Recognition*, vol. 61, pp. 379–393, January 2017.
- [5] B. Xu, B. Chen, J. Wan, H. Liu, and L. Jin, “Target-Aware Recurrent Attentional Network for Radar HRRP Target Recognition,” *Signal Processing*, vol. 155, pp. 268–280, October 2019.
- [6] C. Du, B. Chen, B. Xu, D. Guo, and H.W. Liu, “Factorized discriminative conditional variational auto-encoder for radar HRRP target recognition,” *Signal Processing*, vol. 158, pp. 176–189, January 2019.
- [7] D.P. Kingma, and M. Welling, “Auto-encoding variational Bayes,” *Proceedings of the International Conference on Learning Representations*, May 2014.

- [8] J. An and S. Cho, “Variational autoencoder based anomaly detection using reconstruction probability,” *Special Lecture on IE*, vol. 2, pp. 1–18, February 2015.
- [9] N. Dilokthanakul, Pedro A. M. Mediano, M. Gamelo, Matthew C. H. Lee, H. Salimbeni, K. Arulkumaran, and M. Shanahan, “Deep unsupervised clustering with Gaussian mixture variational autoencoders,” *Proceedings of the International Conference on Learning Representations (ICLR)*, April 2017.
- [10] K. Sohn, X.C. Yan, and H. Lee, “Learning structured output representation using deep conditional generative models,” *Proceeding of the International Conference on Neural Information Processing Systems (NIPS)*, pp. 3483–3491, December 2015.
- [11] Z.J. Wang, B. Chen, H. Zhang, and H.W. Liu, “Variational probabilistic generative framework for single image super-resolution,” *Signal Processing*, vol. 156 pp. 92–105, March 2019.
- [12] S.P. Jacobs and J.A. Osullivan, “Automatic target recognition using sequences of high resolution radar range-profiles,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 36, no. (2) pp. 364–381, April 2000.
- [13] Y. Zhai, B. Chen, H. Zhang, and Z.J. Wang, “Robust variational auto-encoder for radar HRRP target recognition,” *International Conference on Intelligent Science and Big Data Engineering*, pp. 356–367, September 2017.
- [14] L. Du, H. He, L. Zhao, and P.H. Wang, “Noise robust radar HRRP target recognition based on scatterer matching algorithm,” *IEEE Sensors Journal*, vol. 16, no. 6, pp. 1743–1753, March 2016.
- [15] L. Du, H. Liu, and Z. Bao, “Radar HRRP statistical recognition based on hypersphere model,” *Signal Processing*, vol. 88, no. 5, pp. 1176–1190, May 2008.
- [16] P. Vincent, H. Larochelle, I. Lajoie, Y. Bengio, and P. Manzagol, “Stacked denoising autoencoders: learning useful representations in a deep network with a local denoising criterion,” *Journal of Machine Learning Research*, vol. 11 pp. 3371–3408, December 2010.
- [17] G.E. Hinton and R. Salakhutdinov, “Reducing the dimensionality of data with neural networks,” *Science*, vol. 313, no. 504, pp. 504–507, July 2006.
- [18] J. Xie, L. Xu, and E. Chen, “Image denoising and inpainting with deep neural networks,” *Proceedings of the International Conference on Neural Information Processing Systems*, pp. 341–349, December 2012.
- [19] S. Ruder, “An overview of gradient descent optimization algorithms,” *arXiv: 1609.04747*, pp. 1–14, 2017.